

Smart Project & Task Collaboration System

Build a full-stack web application that helps teams manage projects, tasks, team members, and work progress with proper validation and workflow handling.

1. Authentication

User Authentication

- User Signup & Login
- Email + Password authentication
- Secure session/token handling
- After login → redirect to Dashboard
- Include a Demo Login button (pre-filled credentials)

Role-Based Access Control

Support at least:

- Admin
- Project Manager
- Team Member

Role Permissions Example

- Admin → Full system access
- Project Manager → Create/manage projects and assign tasks
- Team Member → Update assigned tasks only

2. Project Management

Users can create and manage projects.

Project Fields

Each project includes:

- Project Name
- Description
- Deadline
- Project Status:
 - Active
 - Completed
 - On Hold

Users can:

- Create projects
- Update project details
- Delete projects
- View all projects

3. Task Management

Users can create tasks under projects.

Task Fields

Each task includes:

- Task Title
- Description
- Assigned Member
- Due Date
- Priority:
 - High
 - Medium
 - Low
- Task Status:
 - Todo
 - In Progress
 - Completed

Features

Users can:

- Create tasks
- Edit tasks
- Delete tasks
- Change task status
- View tasks by project or status

4. Task Validation & Conflict Handling

While creating/updating tasks:

Rules

- Prevent duplicate task titles inside the same project
- Prevent assigning completed tasks
- Prevent setting past dates as deadlines

Example Messages

- “This task already exists in the project.”
- “Completed tasks cannot be reassigned.”
- “Please select a valid deadline.”

5. Team Collaboration

Team Features

Users can:

- Add team members to projects
- Assign tasks to members
- View member-wise task lists

Workload Summary

Display:

- Total tasks per member
- Completed tasks
- Pending tasks

6. Progress Tracking

Dashboard Insights

Display:

- Total Projects
- Total Tasks
- Completed Tasks
- Pending Tasks
- Overdue Tasks

Project Summary Example

- Website Redesign — 5 tasks pending
- Mobile App — 80% completed
- Admin Dashboard — Deadline in 2 days

7. Activity Log

Track recent system activities.

Examples

- 10:00 AM – Project “E-Commerce App” created
- 10:15 AM – Task “Setup API” assigned to John
- 10:30 AM – Task “Homepage Design” marked as Completed
- 11:00 AM – Member added to “Dashboard Project”

Show latest 5-10 activities.

8. Dashboard & Analytics

Display project progress and team productivity insights.

KPI Cards

- Total Projects
- Total Tasks
- Completed Tasks
- Pending Tasks
- Overdue Tasks

Charts / Analytics

Examples:

- Tasks by Priority
- Project Progress Trend
- Team Productivity Overview
- Task Status Distribution

Additional Dashboard Features

- Recent Activities
- Upcoming Deadlines
- High Priority Tasks
- Member Workload Summary

9. Search, Filtering & Productivity Features

Support advanced task and project management features.

Search Features

Users can:

- Search projects by name

- Search tasks by title or description
- Search team members by name

Filter Options

Filter tasks/projects by:

- Project Status
- Task Status
- Priority
- Assigned Member
- Deadline Status (Upcoming / Overdue)

Pagination & Navigation

- Pagination or Infinite Scroll for large datasets
- Responsive table/grid layouts

Sorting Options

Sort by:

- Latest Created
- Nearest Deadline
- Highest Priority
- Recently Updated

Productivity Features

- Quick task status update
- Bulk task actions (optional)
- Task progress indicators
- Recent activity shortcuts

10. Additional Features

- Dark/Light mode
- File attachment support
- Comments on tasks
- Notification system
- Simple analytics chart
- Role-based access (Admin / Member)

11. Technical Instructions

- Any frontend, backend, database, or deployment technology can be used for this assessment.
- Candidates are free to choose any architecture or development approach.
- Clean code structure and proper validation are expected.

12. Deployment Requirements

The application must be deployed and publicly accessible.

Provide:

- Live Application URL
- GitHub Repository
- README Documentation

README should include:

- Project setup instructions
- Features overview
- Environment variables
- Demo credentials
- Deployment instructions